COURSE TITLE : RUBBER TECHNOLOGY
COURSE CODE : 6076
COURSE CATEGORY : E
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 60
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Natural rubber	15
2	Synthetic rubber	15
3	Rubber chemicals and additives	15
4	Different types of production of rubber items	15
TOTAL		60

COURSE OUTCOMES :

SL.NO.	SUB	STUDENT WILL BE ABLE TO
1	1	Understand Natural Rubber.
	2	Know Synthetic Rubber.
	3	Appreciate various rubber chemicals and additives.
2	4	Know different types of production of rubber products.

SPECIFIC OUTCOMES
MODULE- I

1.1.0 Understand the importance, collecting and processing of NR latex

1.1.1 State the importance of NR as a raw material
1.1.2 List out the major sources of NR
1.1.3 Define tapping
1.1.4 List the factors affecting tapping efficiency
1.1.5 Explain the uniqueness of NR
1.1.6 Explain Discuss the possibilities of blending NR with other rubbers
1.1.7 State the applications of NR

1.2.0 Understand the preservation and concentration of NR latex

1.2.1 Define latex
1.2.2 Explain the colloidal nature of latex
1.2.3 Describe the composition of NR latex
1.2.4 State the necessity for preservation of latex
1.2.5 State the requisites of an ideal preservative
1.2.6 Define the terms preservative, short term preservatives and long term preservatives
1.2.7 Explain Discuss the coagulation and method of coagulation of latex
1.2.8 Explain pre-coagulation and use of various anticoagulants
1.2.9 Describe De level Centrifuge with a diagram
1.2.10 State the need for concentration of latex
1.2.11 List the various concentration methods
1.2.12 Explain the principle of creaming of latex

1.2.13 Explain the various creaming agents used (Tamarind seed powder, Sodium alginate, Ammonium alginate etc.) – Definition, examples, their actions, possible mechanisms, their preparation, dosage and method of incorporation
1.2.14 Explain the process of centrifuging of latex with flow chart
1.2.15 Give the meaning of efficiency of centrifuging
1.2.16 State the factors affecting centrifuging
1.2.17 Explain skim latex and skim rubber
1.2.18 Mention the uses of skim rubber

1.3.0 Study the processing of NR latex into dry marketable forms

1.3.1 Define Ribbed smoked sheets (R.S.S.)
1.3.2 Explain the relevance of grading of sheet rubber
1.3.3 Define crepe rubber
1.3.4 Define tyre rubber
1.3.5 Explain the preparation, properties and advantages of tyre rubber

M O D U L E – II

2.1.0 Understand general characteristics of elastomers

2.1.1 Narrate brief history of NR& SR
2.1.2 Classify SR with reference to their applications.
2.1.3 Explain rubber elasticity
2.1.4 Explain general characteristics of SR
2.1.5 Compare elastomers with other polymers like plastics and fibers.
2.1.6 Define elastomers with reference to extensibility and recovery, Glass Transition temperature (Tg) etc.

2.2.0. Understand the structure, preparation and properties of monomers & production
Properties and applications of various general-purpose synthetic rubbers.

2.2.1 Explain the structure, production and properties of the following monomers styrene, butadiene, isoprene, isobutylene, ethylene and propylene.
2.2.2 Explain the structure, production, properties (raw and vulcanisate), curing systems, compounding, processing, grades, trade names and application general purpose synthetic rubbers like
2.2.3 Styrene butadiene rubbers (SBR). Poly butadiene rubber (BR), Isoprene rubber(IR)Butyl (IIR) EPDM
2.2.4 Explain the structure, production, properties (Raw and Vulcanisate), Cure Systems, pounding, processing, grades, trade names and applications of special purpose rubbers
2.2.5 NBR and Discuss the effect of Acrylonitrile, butadiene ratio on the properties of NBR
2.2.6 CR
    2.2.6.1 CSM
    2.2.6.2 Polysulphide rubber
    2.2.6.3 Silicone rubber
    2.2.6.4 Polyurethane rubber

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MODULE-III

3.1.0 Understand principles of compounding

3.1.1 Define compounding and its objective
3.1.2 Prepare a general recipe for product manufacturing
3.1.3 List out the compounding ingredients
3.1.4 Explain base polymer and its function in a compound
3.1.5 Explain vulcanizing agent and its function
3.1.6 State the role of curing system in Recipe
3.1.7 Describe the function of accelerators and their classification
3.1.8 Explain activators and their function and application
3.1.9 Analyze the role of other ingredient such as filler, plasticiser, softener and other special additives used in rubber industry

3.2.0 Comprehend the theory of vulcanization

3.2.1 Explain the changes that occur in a Rubber compound due to vulcanization
3.2.2 Explain sulphur vulcanization
3.2.3 Describe the different types of cross-links and their effect on vulcanisate properties
3.2.4 Explain the theory of vulcanization
3.2.5 Distinguish low sulphur vulcanization and non-sulphur vulcanization with example
3.2.6 Distinguish CV, EV and SEV
3.2.7 Describe the curing system used for NR and SR
3.2.8 Discuss sulphur system and non sulphur system for olefin and non olefin rubbers

3.3.0 Understand preparation, properties & application of compounding ingredients

3.3.1 Explain the classification of fillers.
3.2.2 Describe the preparation and application of various non black fillers like inorganic fillers fibrous filler, organic fillers and resinous filler
3.3.3 Classify carbon black
3.3.4 Describe the methods of manufacturing various types of carbon black
3.3.5 State ASTM D 1765 for grading of carbon black
3.3.6 Summarize the properties of carbon black particle size, structure, chemical and physical nature porosity etc
3.3.7 Describe the procedures for determination of particle size and structure of carbon black
3.3.8 Define degradation
3.3.9 Describe the importance of antidegradants
3.3.10 List the factors affecting degradation
3.3.11 Explain staining and non staining Antioxidants & antiozonants
3.3.12 Explain how plasticizers are classified. Give examples
3.3.13 Explain the effect of plasticizer on Vulcanisate properties
3.3.14 List out the special purpose compounding ingredients used in Rubber compounding.

3.4.0 Understand different vulcanization method other than molding

3.4.1 Explain radiation vulcanization
3.4.2 Explain different methods of batch vulcanization methods
3.4.3 Give examples of product manufactured by batch vulcanization method
3.4.4 Explain continuous vulcanization methods
3.4.5 Explain Roto cure with an example
3.4.6 Explain fluidized bed, microwave curing and give examples of each method
3.4.7 Explain steam tube vulcanization

MODULE-IV

4.1.0 RUBBER PRODUCTS

4.1.1 Name the major rubber products manufacturing industries in India
4.1.2 Compare the per capita consumption of rubber in India and that of other countries
4.1.3 Summarise the statistics relating to the production, consumption and export of dry rubber products.
4.1.4 Classify rubber products according to their application
4.1.5 Mention the different steps involved in the production of dry rubber products

4.2.0 Molded Products

4.2.1 Name important compression molded products
4.2.2 Explain the design, compounding and production details of automobile bushes, mats, paperweights, IB Caps, Door bushes, Nipples, hot water bags
4.2.3 Discuss the design criteria and design of compounds for above products
4.2.4 List hard-rubber products
4.2.5 Explain the chemistry and technology of hard rubber (ebonite) production.
4.2.6 Explain compounding and production of battery box containers

4.3.0 Expanded Products

4.3.1 Define cellular rubber
4.3.2 Classify different cellular rubber
4.3.3 State the principle of creating cellular structure in rubber
4.3.4 Distinguish closed and open cell cellular products

4.4.0. Extruded Products

4.4.1 List the various extruded products
4.4.2 Describe the manufacture of automobile channels
4.4.3 Explain the manufacture of rubber tubing
4.4.4 Discuss the compound design for the above products
4.4.5 Identify major defects and remedies in the manufacturing of the above products

4.5.0 Calendared products

4.5.1 List out various calendared products
4.5.2 Distinguish between supported and unsupported sheets
4.5.3 Describe the manufacture of hospital sheeting and supported sheeting
4.5.4 Explain profile calendaring with examples
4.5.5 Describe the design criteria for various calendared products
4.5.6 Identify the major defects in calendared sheets

COURSE CONTENT

MODULE-I

History & development of natural rubber Major sources – Extraction of Latex — tapping.-Preservation and Concentration of NR latex-Definition of Latex – composition and function of non-rubber constituents, colloidal nature of latex, need for preservation of latex,short and long term preservation – NH3 as ideal preservative, Pre-coagulation and use of anticoagulants – examples.- Coagulation and methods of coagulation of Latex- Concentration of latex – need for concentration of Latex-Latex concentration methods – creaming, centrifuging, evaporation and electro decantation.Centrifuging – principle, machinery, operation on machinery, - importance of centrifuged latex as an industrial raw material and its present trends. - Skim latex and skim rubber – Crepe rubbers – different grades and their processing. - Production requirements-Specialty rubbers -Importance of speciality rubbers in rubber industry--, tyre rubber, powdered natural rubber, -Reclaim Rubber.

MODULE II

Monomers – Preparation and properties of the monomers-styrene, butadiene, isoprene, Isobutylene, ethylene, propylene-structure of Diene monomers-Detailed study of SBR,PBD ,IR,EPDM,& IIR-Monomers – Preparation and properties of the monomers-Acrylonitrile , chloroprene ,-SBR, NBR, Butadiene hydroxy terminated SBR, NBR, Butadiene. –Thermo Plastic Eleastomers-Thermo plastic elastomers, definition, Advantages,modification of elastomers to thermoplastic elastomers. Study of thermoplastic SBR, Ethylene Vinyl acetate, Thermo polyurethane.Blends-Definitions – Advantages, Procedure, Blending for specific properties, study of NR – PBD, NR-SBR-PBD, NR-EVA, NBR-PVC, NR-HSR

MODULE-III

Preparation, Properties & Application of compounding ingredients-Fillers – definition and objectives – classification of fillers – non black fillers preparation, properties and application of non black fillers such as silica, silicates, clays, whiting, lithopone, barites, talc, zinc oxide, MgO, TiO2. Fibrous fillers – asbestos, cellulose fiber, flocks, wood flour -Organic fillers – cork, glue, cyclised NR, hevea -plus, HSR, phenolic resin-Carbon black -Antioxidants and Antiozonents – Plastizers – softeners and extenders -Plasticizer function – classification of plasticizers- special purpose ingredients – blowing agents ,flame retardants ,abrasives , antistatic agents ,integral bonding additives, stiffening agents coupling agents, deodorants etc.-Vulcanization methods-Brief introduction of processing methods for product manufacture - flow chart Moulding – compression – transfer – and injection molding. Compounding for Vulcanizate Properties-General principles of compounding vulcanization properties. Comparison of raw elastomeric properties factors to be considered for designing a rubber compound with examples-Effect of particle size and structure of fillers on processing and vulcanisate properties- Compounding to meet processing requirement viscosity control, control of nerve, adhesion to mill rolls, tack, scorch, calendaring and extrusion – continuous vulcanization.

Calculation of specific gravity and volume costs -with worked out examples

MODULE IV

Molded Products-Introduction to rubber products:Production methods, compound design -Automobile Expanded Products: Types of blowing agents and their comparison- Compounding and Molding of Cellular rubber products, MC, Hawaii sponge –Typical formulations- Extruded Products: Automobile treads, channels, inner tubes, LPG tubes etc- Compound design, Typical formulations, Methods of manufacture. Calendered Products: Supported and unsupported sheeting, Hospital sheeting, Profile calendered products- Design criteria, compound design, typical formulations. Rubber to Metal bonded products-Principle of bonding rubber to metal and textile fibers. Production of rubber covered rolls-metal preparation- compound preparation, assembling curing, finishing-Tank lining- Production method, compounding, compound design.

Production of typewriter rolls, Rice polisher, Textile roller.

REFERENCE

Hand book of natural rubber production	-	Rubber board
High polymer lattices	-	D.C.Blackley
Latex in Industry	-	R.J.Noble